

Advanced Machine Learning

DATA 642

Lecture 8 — Introduction to Tensors

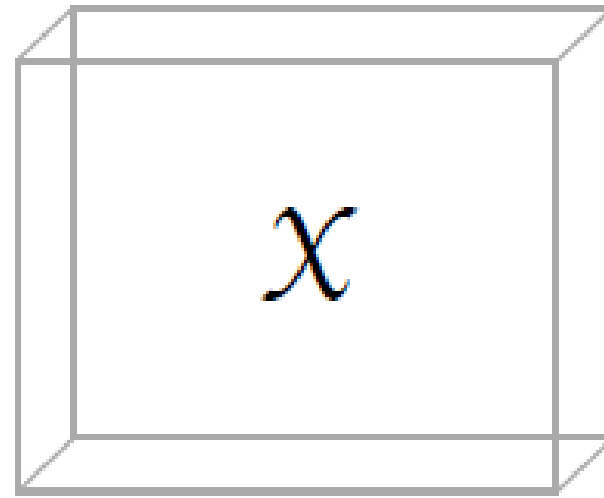
*Figures and notes are from the paper "Introduction to Tensor Decompositions and their Applications in Machine Learning," (<https://arxiv.org/pdf/1711.10781>)

Brief history

- Tensors are generalizations of matrices to higher dimensions and can consequently be treated as multidimensional fields.
- Tensors and their decompositions originally appeared in 1927, but remained untouched by the computer science community until the late 20th century.
- Fueled by increasing computing capacity and a better understanding of multi-linear algebra especially during the last decade, tensors have since expanded to other domains, like statistics, data science, and machine learning.

Tensors Basics

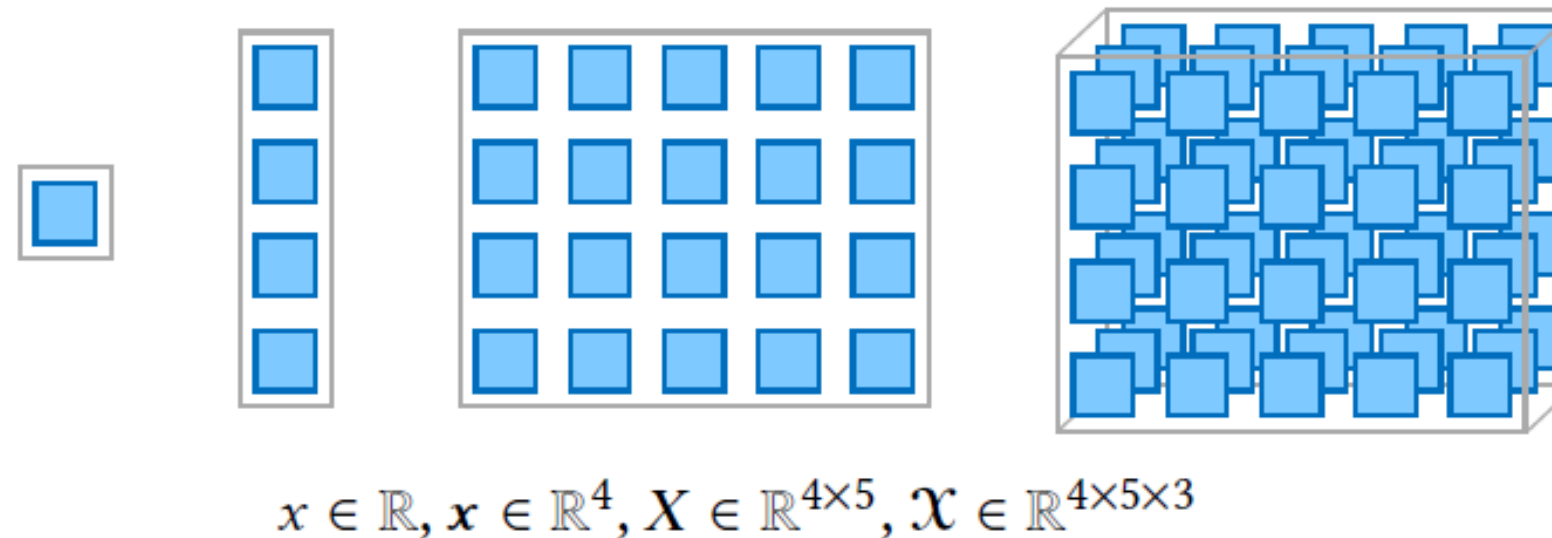
- Tensors can be thought of as multi-way collections of numbers, which typically come from $\mathbb{R}$.
- In the simplest high-dimensional case, such a tensor would be a three-dimensional array, which can be thought of as a data cube.



- In most cases, the notation naturally extends to higher-dimensional tensors.

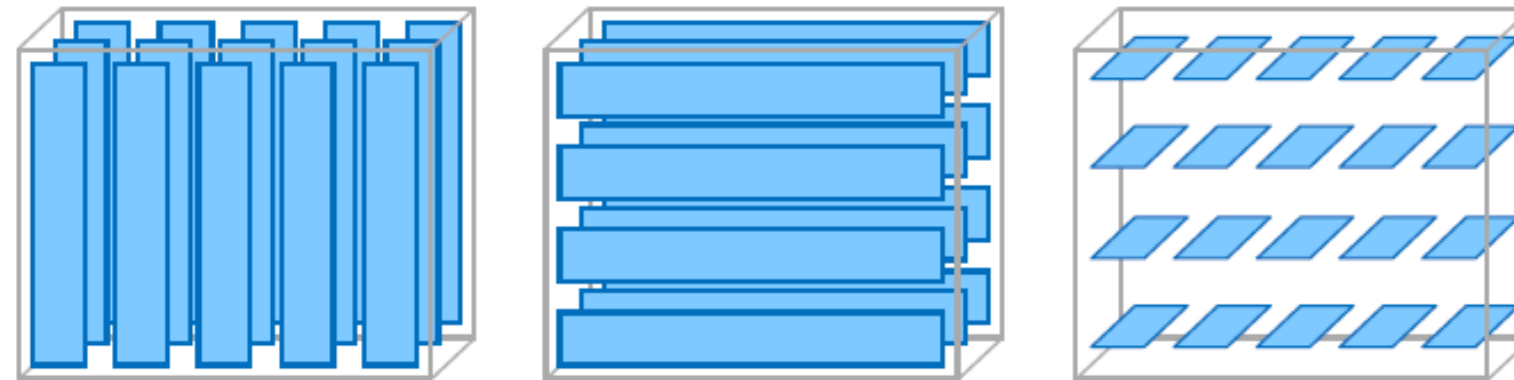
Tensor Order

- The order of a tensor is the number of its dimensions.
- Scalars can therefore be interpreted as zeroth-order tensors, vectors as first-order tensors, and matrices as second-order tensors.
- Notation-wise:
 - Scalars are denoted by lower case letters $x \in \mathbb{R}$.
 - Vectors are denoted by lower case bold letters $\mathbf{x} \in \mathbb{R}^{I_1}$.
 - Matrices are denoted by upper case bold letters $\mathbf{X} \in \mathbb{R}^{I_1 \times I_2}$.
 - Higher-order tensors are denoted by upper case bold Euler script letters $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$.
- The I s denote the number of elements in the respective dimension.



Tensor Indexing

- We can create sub-arrays (or subfields) by fixing some of the given tensor's indices.
- Fibers are created when fixing all but one index.
- Slices (or slabs) are created when fixing all but two indices.
- For a third-order tensor, the fibers are given as $x_{:jk} = x_{jk}$ (column), $x_{i:k}$ (row), and $x_{ij:}$ (tube).
- The slices are given as $\mathcal{X}_{::k} = \mathcal{X}_k$ (frontal), $\mathcal{X}_{:j:}$ (lateral), $\mathcal{X}_{i::}$ (horizontal).



Column, row, and tube fibers of a mode-3 tensor

Outer and Inner Product

- The vector outer product is defined as the product of the vector's elements.
- This operation is denoted by the $\odot$ symbol.
- The vector outer product of two n -sized vectors $\mathbf{a}$ and $\mathbf{b}$ is defined as follows and produces a matrix $\mathbf{X}$:

$$\mathbf{X} = \mathbf{a} \odot \mathbf{b} = \mathbf{a}\mathbf{b}^T \quad (1)$$

- By extending the vector outer product concept to the general tensor product for N vectors, we can produce a tensor $\mathcal{X}$:

$$\mathcal{X} = \mathbf{a}^{(1)} \odot \mathbf{a}^{(2)} \odot \dots \odot \mathbf{a}^{(N)} \quad (2)$$

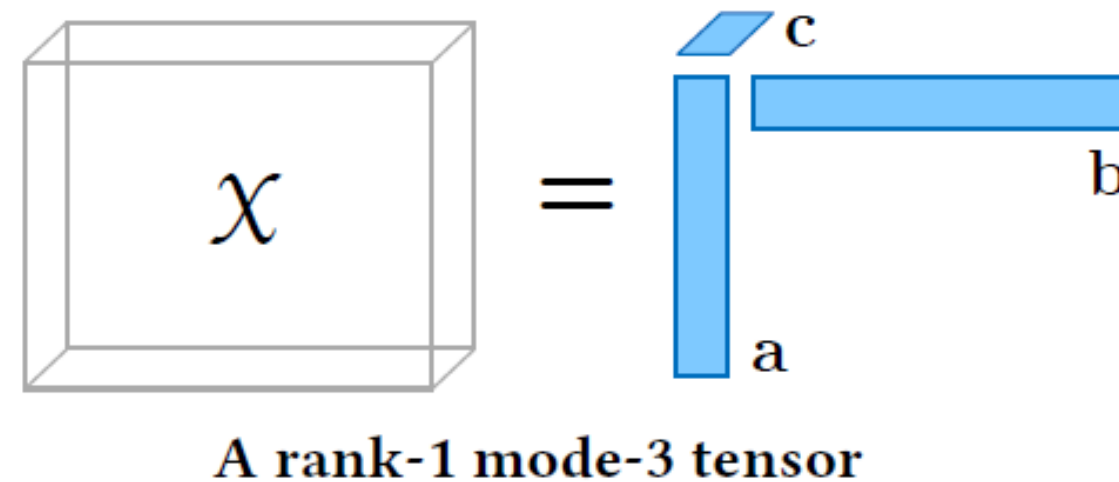
with $x_{i_1 i_2 \dots i_N} = a_{i_1}^{(1)} a_{i_2}^{(2)} \dots a_{i_N}^{(N)}$

- In contrast to the outer product, the inner product of two n -sized vectors $\mathbf{a}$ and $\mathbf{b}$ is defined as:

$$x = \langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a}^T \mathbf{b} = \sum_{i=1}^n a_i b_i = a_1 b_1 + a_2 b_2 + \dots + a_n b_n \quad (3)$$

Rank-1 Tensors

- An N -way tensor is of rank-1 if it can be strictly decomposed into the outer product of N vectors.
- Intuitively, this means that we introduce different scalings of a sub-tensor as we add more dimensions when building up the complete tensor.
- A rank-one matrix can therefore be written as $\mathbf{X} = \mathbf{a} \odot \mathbf{b}$ and a rank-one 3-way tensor as $\mathcal{X} = \mathbf{a} \odot \mathbf{b} \odot \mathbf{c}$.
- The general N -way form was already introduced in the previous equation. A graphical view of the rank-1 concept is given in the following figure



Tensor Rank

- The rank of a tensor $\text{rank}(\mathcal{X}) = R$ is defined as the minimum number of rank-one tensors needed to produce $\mathcal{X}$ as their sum.
- The general N -way form is given as:

$$\mathcal{X} = \sum_{r=1}^R \lambda_r \mathbf{a}_r^{(1)} \odot \mathbf{a}_r^{(2)} \odot \cdots \odot \mathbf{a}_r^{(N)} \quad (4)$$

- The $\mathbf{A}$ s are called factor matrices and hold the combination of the vectors from the rank-one components as columns. A specific $\mathbf{A}$ therefore has the form $\mathbf{A} = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_R]$.
- We introduced an additional factor λ_r which is often used to absorb the respective weights during normalization of the factor matrices' columns. This usually means normalizing the sum of the squares of the elements in each column to one. Note that $\lambda \in \mathbb{R}^R$.